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RESEARCH ARTICLE

Study on Real-Time Battery Temperature Prediction Based on Coupling of Multiphysics Fields and Temporal Networks

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ABSTRACT Real-time temperature prediction is essential for ensuring the thermal safety of Lithium-ion batteries (LIBs), yet its industrial application faces challenges due to fluctuations in operating conditions such as temperature, voltage range, capacity degradation, and current rates (C-rates). To address this, we introduce a novel framework, Transformer-GPR, which merges the physical battery model with a Transformer-based network. This integration facilitates the offline training of hyperparameters, enhancing real-time temperature prediction accuracy. Additionally, we employ two residual models using Gaussian Process Regression (GPR) to correct for local temperature deviations. The Transformer-GPR framework is designed to predict temperature accurately across the entire lifecycle of LIBs with limited data and under varied operational conditions. It has been benchmarked against several existing methods, showing superior interpretability, accuracy, and transferability. Validation with operational data from a pure electric vehicle confirmed the model's efficacy; it precisely predicted temperature change sequences, with an RMSE of 0.048, an MAE of 0.036, and a maximum error of 0.28, using training inputs from similar vehicles.

INDEX TERMS Lithium-ion battery, temperature prediction, battery modeling, neural network, Gaussian process regression.

I. INTRODUCTION

As the continuous consumption of global fossil fuels advances, Electric Vehicles (EVs) have gained widespread application due to their zero carbon emission characteristic and potential as a substitute for traditional fuel-driven vehicles [1]. Among these, Battery Electric Vehicles (BEVs) based on Lithium-ion battery systems have emerged as the most popular choice due to their high power and energy density, long cycle life, and low self-discharge rate compared to other common battery technologies [2]. However, the performance of LIBs is greatly influenced by operating temperature. Inappropriate operating temperatures limit performance and accelerate degradation [3]. Moreover, when operating temperatures reach a threshold, Thermal Runaway (TR) is triggered, leading to severe vehicle accidents [4]. Therefore, real-time temperature prediction of lithium-ion batteries is crucial for ensuring safety and enhancing potential lifespan [5].

Typically, methods for measuring battery temperature can be categorized into experimental and model-based approaches [6]. Experimental methods involve using sensors to measure and display the temperature distribution across the entire battery, including internal and surface temperatures. Hussein and Fardoun [7] introduced an adaptive sensorless measurement technique for internal temperatures of lithium-ion batteries, with the proposed technique identifying the zero-cross frequency where the impedance phase is zero. Through this zero-cross frequency value, the internal temperature of the battery is estimated. Experiments were conducted using a Samsung 3.8V/2600mAh lithium-ion battery within a temperature range of −20°C to +50°C for validation.

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Zhang et al. [8] developed an estimation method for the total heat generation of lithium-ion batteries based on Dual Temperature Measurement (DTM) and a dual-state thermal model. The results demonstrated that the algorithm could track the designed multi-stage heat generation profile and the rate of heat generation during conventional rapid charging processes in real-time. Zhu et al. [9] proposed integrating multi-point thin-film sensors into lithium-ion battery electrodes, and the results indicated that the thin-film sensors, possessing electrochemical inertness, could display internal temperature changes under different current rates in real-time. Over 100 cycles, the integrated battery maintained stable capacity and high coulombic efficiency. Xiaoli et al. [10] introduced a new method for estimating the total heat generation of a cylindrical 21700 battery at a 1C discharge rate under ambient room conditions, which was approximately 3.2 kilojoules, with an average heat generation rate of about 0.9 watts. Similarly, when the discharge rate increased to 2 C, the total heat generation rose to approximately 5.0 kilojoules, with the average heat generation rate increasing to about 2.8 watts. These results suggest that the proposed measurement methods can closely approximate the actual operating state of the battery. Du et al. [11] proposed a sensorless temperature estimation method based on broadband impedance spectroscopy, utilizing a Pseudo-Random Sequence (PRS) with limited signal level for impedance measurement. Experimental results indicated good estimation accuracy under various conditions. However, due to high costs, insufficient safety, and operational complexity, the feasibility of experimental methods is relatively low.

Therefore, neural networks represent an appropriate solution for effectively modeling the complex and nonlinear relationships between input and output values in lithium-ion battery applications [12]. Characterized by real-time capabilities and low computational times, they facilitate the understanding of the overall correlations between different process parameters without the need for detailed modeling of the physicochemical interactions [13]. Several methods based on Artificial Neural Networks (ANNs), with varying inputs and target values, have been introduced in the literature for thermal modeling of lithium-ion batteries [14]. Arora et al. [15] employed an ANN with a feed-forward (FF) topology to predict the heat generation of various pouch batteries within an 8-20 Ah range under different operating conditions. Jan et al. [16] developed and parameterized a novel Nonlinear Autoregressive with Exogenous inputs (NARX) network for a large 25Ah prismatic battery cell. This NARX network was compared with a feedforward network using the same general structure and input data, and both artificial neural network approaches demonstrated their accuracy in long-term temperature prediction over a 10-hour period, with precision within 1K. Wang et al. [17] proposed a new framework named Battery Informed Neural Network (BINN), integrating the physical battery model into a network based on Long Short-Term Memory (LSTM), which can be trained end-to-end for battery temperature prediction. The results indicated that BINN possesses interpretability and offers better generalization and transferability compared to traditional learning-based methods.

Although traditional RNN and LSTM prediction methods theoretically can handle long-distance dependencies, in practice, they are affected by vanishing or exploding gradients, significantly reducing this capability [18]. The Transformer is distinguished by its efficient parallel processing ability, strong long-distance dependency capturing capacity, and flexible attention mechanism [19]. Moreover, it can be integrated with big data, neural networks, and the Internet of Things for lifecycle condition monitoring and assisting in decision-making. Previous studies have applied Transformer technology to predict the State of Charge (SOC) and capacity degradation of lithium-ion batteries. Shi et al. [20] designed a specialized Transformer-based network architecture that uses time-resolved battery data (i.e., current, voltage, and temperature) as inputs to estimate SOC, with the best model exhibiting a Root Mean Square Error (RMSE) of less than 0.5 on the test dataset, accurately estimating SOC under a wide range of operating profiles in dynamic running and aging conditions. Chen et al. [21] developed a Transformer-based neural network for Remaining Useful Life (RUL) prediction, which captures temporal information and learns useful features by feeding the reconstructed sequences into the Transformer network. The results indicated that their method performs well in predicting RUL. Previous studies have demonstrated the feasibility of Transformer technology in Battery Thermal Management Systems (BTMS). However, there has been little comprehensive research on real-time battery temperature prediction using Transformer models. Therefore, this study proposes a method based on the Transformer-GPR neural network for real-time temperature prediction and pattern analysis.

The main contributions of this work are as follows:

1) An equivalent circuit-based coupled model was established to facilitate the selection and calculation of input feature factors.
2) An innovative framework was developed for real-time lithium-ion battery temperature prediction, which enables accurate temperature forecasting with limited measurement parameters. The framework's performance was validated through comparisons with various existing methods.
3) A Transformer model, complemented by a Gaussian Process Regression (GPR) residual model, was employed to achieve precise battery temperature predictions. The Transformer model effectively captured the overall trend of battery temperature changes, while the GPR model provided corrections for local temperature variations.
4) An experimental platform was constructed to assess the generalizability of the model, ensuring the accuracy and reliability of the prediction framework.

II. METHODOLOGY

A. ESTABLISHMENT OF THE BATTERY PHYSICAL MODEL

In examining the key factors that influence battery performance and safety, the electro-thermal coupling phenomenon is pivotal due to its direct impact on battery temperature stability [22] This paper presents a comprehensive approach to deeply understand and accurately predict temperature variations in batteries under practical conditions. Our approach integrates electrical, thermal generation, and temperature models to precisely determine the internal core temperature of the battery cell. This parameter is essential not only for revealing the battery's critical thermal behavior but also for designing effective Battery Thermal Management Systems (BTMS).

Our methodology constructs thermal models within an electrochemical and electro-thermal coupling framework to enhance both the accuracy and practicality of predictions. The integrated model framework comprises electrical, thermal generation, and temperature sub-models that function as a dynamic interactive system. Utilizing a collaborative simulation approach, this framework dynamically captures the interplay between electrical and thermal parameters. It continuously updates the battery cell's parameters in real time by feeding back temperature data into the electrical and thermal generation models at each simulation step.

Next, the electrical model of the battery is investigated. The electrical model used in this study is a second-order RC equivalent circuit model [23], consisting of a constant voltage source, an internal resistance, and two pairs of capacitor resistors, as shown in Figure 1. The constant voltage source is used to describe the relationship between the battery's open-circuit voltage (OCV) and state of charge (SOC), while the two pairs of capacitor resistors simulate the polarization phenomenon during battery charging and discharging processes. According to Kirchhoff's voltage and current laws, the state variables of the equivalent circuit model can be described by the following equations.

$$\frac{dSOC\,(t)}{dt} = -\frac{I\,(t)}{3600C_n} \tag{1}$$

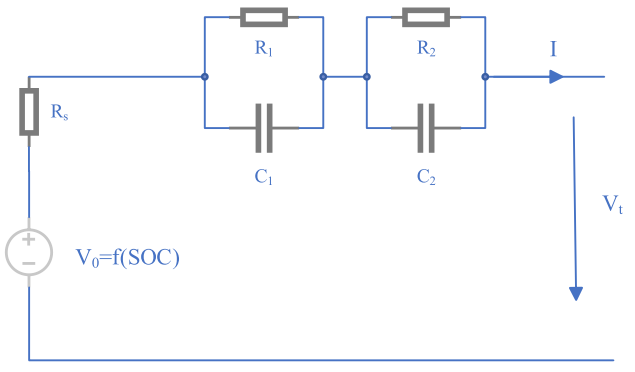


FIGURE 1. Battery second-order RC circuit model.

$$\frac{dV_1\,(t)}{dt} = -\frac{V_1\,(t)}{R_1\,(t)\,C_1\,(t)} + \frac{I\,(t)}{C_1\,(t)} \tag{2}$$

$$\frac{dV_2\,(t)}{dt} = -\frac{V_2\,(t)}{R_2\,(t)\,C_2\,(t)} + \frac{I\,(t)}{C_2\,(t)} \tag{3}$$

where I represents the operating current, C_n denotes the rated capacity, and V_1 and V_2 represent polarization voltages. Parameters R_1, C_1, R_2, and C_2 respectively represent the two pairs of capacitor resistors.

The battery terminal voltage can be calculated from the above state variables, as shown in Equation 4.

$$V_t\,(t) = V_0\,(SOC\,(t)) - V_1\,(t) - V_2\,(t) - R_sI\,(t) \tag{4}$$

where V_0 represents the open-circuit potential, and R_s denotes the internal resistance.

The model parameters are identified using batch least squares method. Let $U_L = V_t - V_0$. The transfer function of the second-order RC equivalent circuit model can be obtained from Equations 2 to 4 as follows:

$$G = \frac{U_L\,(s)}{I\,(s)} = -\left(R_s + \frac{R_1}{1 + R_1C_1s} + \frac{R_2}{1 + R_2C_2s}\right) \tag{5}$$

Using the bilinear transform, the transfer function is converted from the s-domain to the z-domain, with a sampling frequency set at 1 Hz, resulting in the system's discretized transfer function as shown in Equation 6.

$$G\left(z^{-1}\right) = \frac{b_3 + b_4z^{-1} + b_5z^{-2}}{1 - b_1z^{-1} - b_2z^{-2}} \tag{6}$$

$$\begin{cases} R_s = \dfrac{b_3 - b_4 + b_5}{1 + b_1 - b_2} \\ R_1R_2C_1C_2 = \dfrac{1}{4}\cdot\dfrac{1+b_1-b_2}{1-b_1-b_2} \\ R_1C_1 + R_2C_2 = \dfrac{1 + b_2}{1 - b_1 - b_2} \\ R_s + R_1 + R_2 = \dfrac{b_3 + b_4 + b_5}{1 - b_1 - b_2} \\ (R_s + R_2)\,R_1C_1 + (R_s + R_1)\,R_2C_2 = \dfrac{b_3 - b_5}{1 - b_1 - b_2} \end{cases} \tag{7}$$

The discretized transfer function of the system allows for the derivation of the model's discrete difference equation, as shown in Equation 8.

$$\begin{aligned} U_L\,(k) = {} & b_1U_L\,(k-1) + b_2U_L\,(k-2) + b_3I\,(k) \\ & + b_4I\,(k-1) + b_5I\,(k-2) \end{aligned} \tag{8}$$

After N+2 sampling instants, the following set of input-output data relationship is obtained.

$$\begin{cases} U_L\,(3) = b_1U_L\,(2) + b_2U_L\,(1) + b_3I\,(3) \\ \quad + b_4I\,(2) + b_5I\,(1) \\ U_L\,(4) = b_1U_L\,(3) + b_2U_L\,(2) + b_3I\,(4) \\ \quad + b_4I\,(3) + b_5I\,(2) \\ \vdots \\ U_L\,(N+2) = b_1U_L\,(N+1) + b_2U_L\,(N) + b_3I\,(N+2) \\ \quad + b_4I\,(N+1) + b_5I\,(N) \end{cases} \tag{9}$$

Then, based on Equation 7 and the batch least squares formula 10, the identification parameters R_s, R_1, C_1, R_2, and C_2 are finally computed.

$$B_N = \left(\theta_N^T \cdot \theta_N\right)^{-1} \cdot \theta_N^T \cdot Y_N \tag{10}$$

In the equation, according to Equation 9, B, θ, and Y respectively represent the following:

$$B_N^T = (b_1, b_2, b_3, b_4, b_5) \tag{11}$$

$$\theta_N^T = \begin{pmatrix} U_L(2)\, U_L(3) \cdots U_L(N+1) \\ U_L(1)\, U_L(2) \cdots U_L(N) \\ I(3)\, I(4) \cdots I(N+2) \\ I(2)\, I(3) \cdots I(N+1) \\ I(1)\, I(2) \cdots I(N) \end{pmatrix} \tag{12}$$

$$Y_N^T = (U_L(3), U_L(4), \cdots, U_L(N+2)) \tag{13}$$

B. BATTERY HEAT GENERATION MODEL

According to the working principle of lithium-ion batteries, the charging and discharging processes involve the movement of lithium ions and electrons in the internal and external circuits, respectively, generating heat at each step. From an electrochemical perspective, the heat generation in lithium-ion batteries mainly originates from four sources: reaction heat, Joule heating, polarization heat, and side reaction heat. Currently, both domestic and international studies widely adopt the battery heat generation rate model established by Bernardi et al. [24], as shown in Equation 14.

$$Q = I\left[(U_{OCV} - U) + T_b \frac{\partial U_{OCV}}{\partial T_b}\right] \tag{14}$$

In the equation: U_{OCV} represents the open circuit voltage, measured in volts (V); U denotes the operating voltage, measured in volts (V); T_b stands for the battery's thermodynamic temperature, measured in Kelvin (K); while $\frac{\partial U_{OCV}}{\partial T_b}$ represents the entropy coefficient, measured in volts per Kelvin (V/K), which depends on the SOC and significantly influences the temperature behavior of the model.

The thermal model used considers the core temperature and surface temperature of the cylindrical cell as two state variables. Based on the law of energy conservation, the heat exchange relationship between them and the environment is established, as shown in Figure 2. The equations describing the state variables of the thermal model can be provided by Equations 15 and 16.

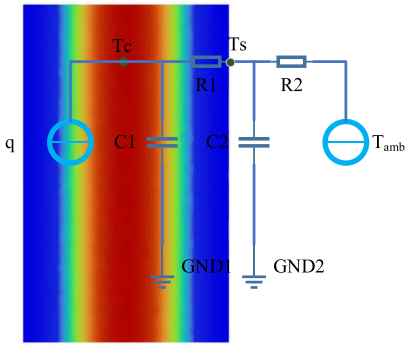


FIGURE 2. Battery thermal model diagram.

$$\frac{dT_s(t)}{dt} = \frac{T_{amb} - T_s(t)}{R_u C_s} - \frac{T_s(t) - T_c(t)}{R_c C_s} \tag{15}$$

$$\frac{dT_c(t)}{dt} = \frac{T_s(t) - T_c(t)}{R_c C_c} - \frac{Q(t)}{C_c} \tag{16}$$

In the equations, T_s and T_c represent the surface temperature and core temperature of the battery, respectively. T_{amb} represents the ambient temperature, while R_u, R_c, C_c, and C_s represent the convective thermal resistance, conductive thermal resistance, core heat capacity, and surface heat capacity, respectively. The heat generation $Q(t)$ can be calculated using Equation 17.

$$Q(t) = I(t) \cdot [V_1(t) + V_2(t) + R_s \cdot I(t)] \tag{17}$$

This study presents a comprehensive physical battery model that precisely tracks the process from heat generation to temperature change. Based on the key factors provided by the electrical model of the battery and incorporating the mechanisms of heat generation and heat transfer described in the thermal model, this model elaborately outlines the electrical, thermal, and heat transfer behaviors of the battery during operation. Notably, surface temperature observations of the battery, time series information, and aging status have been employed as input features $x(t)$ for the Neural Network analysis, as shown in Equation 18. This approach allows for a granular understanding and prediction of battery performance, which is critical for optimizing battery design and management.

$$x(t) = \left[T_c', U, I, SOC, T_s, Q, C, N_{cycle}\right]^\tau \in \mathbb{R}^8 \times \mathbb{N} \tag{18}$$

The output y of the neural network is defined by Equation 19 and serves as the input for the next step.

$$y(t) = T_c \tag{19}$$

The input features $x(t)$ for the neural network include experiment time t, cell voltage U, cell current I, state of charge (SOC), surface temperature T_c, maximum discharge capacity of the last cycle C, number of cycles N_{cycle}, the previous temperature of the cell T_s', and the overall heat generation of the cell Q.

III. BATTERY ONLINE TEMPERATURE PREDICTION

A. GENERAL SETUP

A novel modeling method based on neural networks has been employed for predicting the thermal behavior of lithium-ion batteries. The proposed framework for battery temperature prediction is illustrated in Figure 3, encompassing both offline training and online application processes. A Transformer model, equipped with an attention mechanism, is used to predict future operating temperatures, and two residual models based on Gaussian Process Regression (GPR) are employed to compensate for prediction errors caused by environmental temperatures and battery degradation.

In the Transformer model, activation functions are added at different layers to impart nonlinear effects on the model's predictive outcomes and learning capabilities. In this study, the feedforward network of the Transformer model utilizes the commonly used ReLU function as the activation function:

$$f(u) = ReLU(u) = max(0,u) \tag{20}$$

Other activation functions, such as the sigmoid, have been tested, but the ReLU function has shown the best results for the current application. Weights and biases are determined during the training phase. In this process, Mean Squared Error (MSE) is employed as the loss function for the output, with several algorithms available to minimize this error function. For instance, the gradient descent method optimizes weights along the steepest negative gradient. Another method is Newton's method, which adjusts weights using a search method close to the minimum error. In the current distribution, the Levenberg-Marquardt algorithm (LM) has been applied, which is considered a combination of the Gauss-Newton algorithm and the gradient descent method. As a result, this algorithm has been identified in the literature as having the fastest convergence rate. The LM algorithm minimizes the objective function by iteratively updating the parameter vector $\emptyset$, with the update formula given as:

$$\emptyset_{new} = \emptyset + \left(\mathcal{J}^T\mathcal{J} + \lambda D\right)^{-1}\mathcal{J}^T r \tag{21}$$

where $\emptyset$ is the parameter vector; J is the Jacobian matrix of the objective function with respect to the parameters; r is the residual vector, representing the difference between observed and model-predicted values; λ is a non-negative regularization parameter that controls the algorithm's inclination towards gradient descent or Gauss-Newton; and D is a diagonal matrix, typically composed of the diagonal elements of $\mathcal{J}^T\mathcal{J}$.

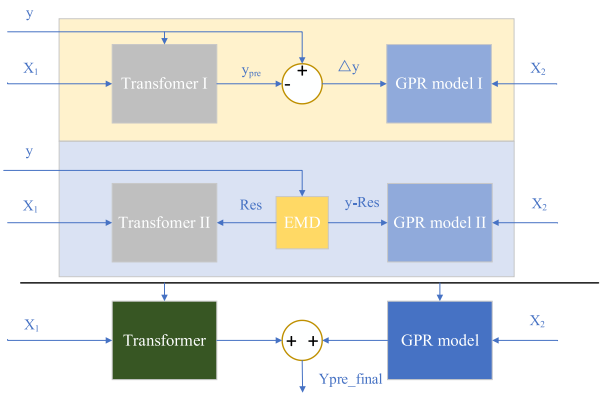


FIGURE 3. The framework of battery temperature prediction.

Data input requires preprocessing, which consists of two parts: the first step is data smoothing to eliminate anomalies. In this paper, Kalman filtering is employed for noise reduction to enhance data smoothness. The second step involves normalization to mitigate the impact of variations in the input data distribution on the network model. Min-max normalization is utilized for data processing in this study. Let $x_t = \{x_t(1), x_t(2), \ldots, x_t(n)$ represent the input sequence of length n, where x_t is mapped to the interval (0, 1].

The normalization is expressed as:

$$x_t' = \frac{x_t - min(x)}{max(x) - min(x)} \tag{22}$$

In this formula, x_t denotes the temperature-related monitoring data of the equipment at time $min(x)$ and $max(x)$ are the minimum and maximum values of the monitored data for the equipment at all times, respectively, and x_t' represents the normalized data.

B. TRANSFORMER

The Transformer is a deep learning model specifically designed for processing sequential data and has demonstrated exceptional performance, particularly in the field of Natural Language Processing (NLP). It was first introduced by Ashish Vaswani and others in 2017 [25]. The core idea of the Transformer model lies in leveraging the self-attention mechanism to process sequence data. This mechanism enables the model to directly focus on any part of the input sequence, thereby capturing long-distance dependencies within the sequence more effectively.

The standard Transformer network is a sequence-to-sequence structure comprised of an encoder and a decoder. The input sequence of the Transformer network is mapped by the encoder to a higher-dimensional vector, which is further processed by the decoder to generate an output sequence. Unlike RNN networks, the Transformer network establishes long-term dependencies through the multi-head attention mechanism. This paper utilizes the encoder of the Transformer network to capture the long-term dependencies of temperature from the monitoring data of batteries. A fully connected layer is then used to make predictions based on the encoder's output, yielding the network's output results.

In this study, a Transformer encoder is utilized to learn the long-term dependencies of temperature variations from historical battery operation data. The Transformer layer consists of two sub-layers: the multi-head attention layer and the feedforward fully connected layer. To fully consider the positional information of sequences, basic positional encoding is applied to annotate the sequences with relative positional markings. Different frequencies of sine and cosine functions are employed for positional encoding.

$$\begin{cases} P_j^{2k} = sin\left(j/10000^{2k/d_{model}}\right) \\ P_j^{2k+1} = cos\left(j/10000^{2k/d_{model}}\right) \end{cases} \tag{23}$$

$$X_t = x_t' + P_j \tag{24}$$

In the equation, P_j represents the positional encoding for time step j, d_{model} is the dimension of the input sequence x_t', and X_t is the output sequence after positional encoding processing.

The purpose of the multi-head attention layer is to capture dependencies between features. This layer divides the output

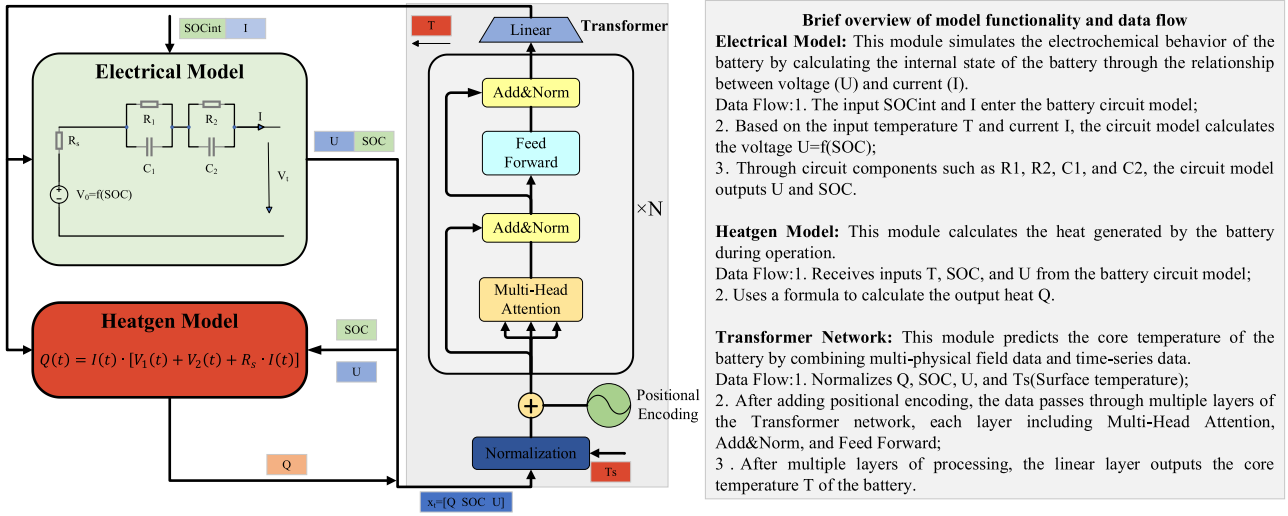


FIGURE 4. Predictive model schematic diagram.

sequence X_t from the positional encoding layer into multiple groups, and processes each group independently using self-attention mechanisms. Within each group, the input sequence is transformed into three vectors: query, key, and value vectors. Subsequently, attention scores are computed between the query and key vectors, which are then used to weight the value vectors. The weighted value vectors are further concatenated and transformed to generate the output of the attention mechanism. Finally, the results of each self-attention mechanism in each group are concatenated and subjected to a linear transformation to obtain the output sequence. The expression for the multi-head attention mechanism of the (L-1)-th layer is provided.

$$\begin{cases} Q = X_t W^Q \\ K = X_t W^K \\ V = X_t W^V \end{cases} \tag{25}$$

$$head_i = Attention\left(Q, K, V\right) \tag{26}$$

$$Attention\left(Q, K, V\right) = softmax\left(\frac{QK}{\sqrt{d_k}}\right)V \tag{27}$$

$$MultiHead\left(H^{L-1}\right) = \left[head_1; head_2; \cdots; head_h\right] W^O \tag{28}$$

In the equation, Q represents the query vector, K is the key vector, V is the value vector, $W^{(\cdot)}$ represents different weight values, H^{L-1} denotes the attention function for the (L-1)th layer. Here, $W_i^Q \in R^{d_{model} \times d_k}$, $W_i^K \in R^{d_{model} \times d_k}$, $W_i^V \in R^{d_{model} \times d_v}$, $W_i^O \in R^{d_{model} \times hd_v}$ are trainable weights, h represents the number of heads in multi-head attention mechanism, and $d_k = d_v = d_{model}/h$ to avoid vanishing gradients.

The feed-forward fully connected layer consists of two linear transformations with an intermediate ReLU activation function. The addition of the feed-forward fully connected layer enhances the model's ability to learn complex nonlinear relationships between input sequences. Nonlinear activation functions allow the layer to learn interactions between input features, while linear transformations enable the layer to learn different weights for different positions in the input sequence. To prevent the attention mechanism from fitting complex processes inadequately, the inclusion of the feed-forward network increases nonlinear expressive power, thereby enhancing the model's capacity. This capacity is derived from the interaction with the previous multi-head attention layer.

$$H^L = FFN\left(MultiHead\left(H^{L-1}\right)\right) \tag{29}$$

$$FFN\left(x\right) = ReLU\left(xW_1 + b_1\right)W_2 + b_2 \tag{30}$$

To predict the unknown temperature, a fully connected layer is used to map the representation learned from the last Transformer unit to the final prediction y_t.The overall prediction process is shown in Figure 4.

$$y_t = f\left(W_p H^h + b_p\right) \tag{31}$$

Here, W_p, b_p, H^h, and f represent the weights, bias, input, and mapping function of the prediction layer, respectively.

The primary objective of thermal modeling methods is to ascertain the core temperature of battery cells, capturing their critical thermal behavior essential for an effective Battery Thermal Management System (BTMS). As shown in Figure 4, our approach integrates thermal models within an electrochemical/electro-thermal framework, encompassing three subsystems: the electrical model, the heat generation model, and the thermal model. This framework receives external inputs such as the battery's initial State of Charge (SOC_{int}) and current (I). Co-simulation methods are employed to facilitate interactions between electrical and thermal parameters, allowing for dynamic updates of battery characteristics with each time step.Our study employs

a Transformer-based modeling approach for predicting the thermal behavior of prismatic lithium-ion battery cells. This architecture features a self-attention mechanism that calculates dependencies across all sequence elements, enhancing the ability to capture complex, long-range interactions in time series data beyond traditional models. The Transformer encoder leverages self-attention to weigh the significance of different time steps, prioritizing historical data crucial for current predictions. This significantly improves pattern recognition across various time scales. Moreover, the encoder's hierarchical structure extracts multi-scale features vital for accurately tracking dynamic changes in the data.

The neural network depicted in Figure 4 employs the ReLU activation function in each encoding layer's Feed-forward Neural Network (FFN), although other functions like tanh were tested. ReLU has proven most effective for this application. A linear function is applied in the output layer, with weights and biases optimized during training. The Mean Squared Error (MSE) serves as the loss function. We utilize several algorithms to minimize this error, including gradient descent, which optimizes weights along the steepest negative gradient. The Levenberg-Marquardt algorithm, a hybrid of gradient descent and Newton's method, is also applied, offering the fastest convergence according to existing literature [26].

The core of the Transformer-GPR framework is its architecture based on the self-attention mechanism of the Transformer, which excels at processing long sequence data but also introduces significant computational complexity and resource consumption. Specifically, the computational complexity of the Transformer model is $O(n^2)$, where n represents the length of the input sequence. This complexity necessitates substantial computational resources, such as GPU memory and processing power, particularly when handling large datasets or extended time series. To address these challenges, this paper employs techniques such as parameter quantization and knowledge distillation to reduce the computational load and enhance the operational efficiency of the model.

C. RESIDUAL MODEL BASED ON GPR

In electric vehicle applications, the accurate prediction of battery temperature is essential for optimizing performance and extending battery life. Battery temperature is heavily influenced by environmental conditions, state of charge, discharge rates, and degradation. Fluctuations in ambient temperature can lead to significant variations in internal battery temperature, impacting efficiency, energy storage capacity, and degradation rates. To investigate the effects of lithium battery capacity degradation and ambient temperature on battery temperature variations, the parameters of the 18650 lithium battery, as shown in Table 1, were used. An experimental setup was established using a Neware BTS series high-precision battery testing system (CT-4008T), an RYHS100/3-40AB multi-layer constant temperature and humidity test chamber, and a computer interface to control

TABLE 1. Battery specification sheet.

Parameter	Value	Parameter	Value
Nominal Capacity	3Ah	Nominal Voltage	3.6V
Charging Voltage	4.2±0.05V	Charging Cut-off current	4.0±0.5A
Maximum Charging Current	1.0A (0℃～+12℃) 4.0A (12℃～+45℃)	Discharging Cut-off voltage	2.5V
Maximum Discharging Current	50A 18A(Continuous discharge)	Operating Temperature	0℃～45℃ (charging) -20℃～60℃ (discharging)

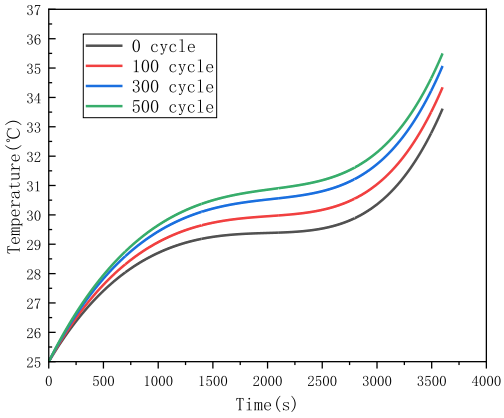


FIGURE 5. The change in temperature with the number of cycles.

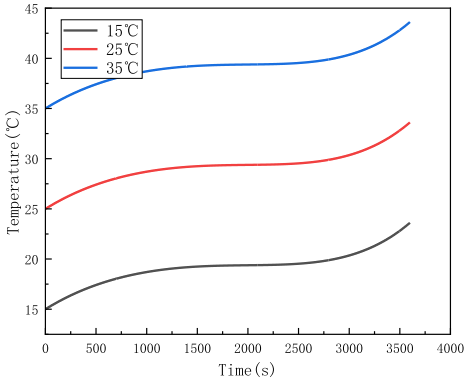


FIGURE 6. The change in temperature with the ambient temperature.

experimental conditions, including ambient temperature, charge-discharge rates, and cut-off voltages. The temperature sampling frequency was maintained at 1 Hz. The experimental results, depicted in Figures 5 and 6, demonstrate that battery temperature varies under different ambient temperatures and cycle numbers at the same discharge rate.

Previous studies highlight a complex interplay between a battery's internal temperature, ambient conditions, and its aging process. During periods of low ambient temperatures, internal temperatures drop, potentially reducing efficiency and accelerating capacity loss. Conversely, higher ambient temperatures may increase internal temperatures, which can

restore efficiency but also risk overheating and expedited degradation unless carefully managed.

To predict battery temperature accurately under dynamic conditions, it is crucial to account for the direct impact of ambient temperature and the indirect effects of battery capacity degradation. While Transformer models effectively capture general temperature trends, they often struggle with subtle, localized fluctuations influenced by calendar time, ambient temperature, and battery degradation.

To address this limitation and improve prediction accuracy, integrating residual models with Gaussian Process Regression (GPR) has proven effective. GPR's capability to probabilistically model complex nonlinear relationships enables the refinement of predictions by accounting for residual errors potentially overlooked by Transformer models.

Gaussian Process Regression (GPR) is a powerful nonlinear regression method, widely used in machine learning and statistics to model and predict complex data behaviors. As a form of Bayesian regression, its core concept involves modeling the regression problem as a Gaussian process—a stochastic process defined over a continuous input space where the joint distribution of any set of points follows a Gaussian distribution. For typical regression problems, considering Gaussian white noise, the observational data can be expressed as follows:

$$y\left(x_i\right) = f\left(x_i\right) + \epsilon \tag{32}$$

where $y(x)$ is the observed value at point x, $f(x) = [f(x_1), f(x_2), \cdots, f(x_n)]$ is a Gaussian process, x_i is the i-th feature, ϵ is the noise added to each observation, $\epsilon \sim \mathbb{N}\left(0, \sigma_n^2\right)$, and $f(x)$ can be described as $f(x) \sim N(0, K)$, where $K_{ij} = k(x_i, x_j)$ is the kernel function used to measure the distance between points x_i and x_j, primarily determining the estimation accuracy of the regression model. The most commonly used kernel function is the squared exponential function, which can be represented as:

$$k\left(x_i, x_j\right) = \sigma_f^2 exp\left(\frac{-\left\|x_i - x_j\right\|^2}{2l^2}\right) \tag{33}$$

In the Gaussian Process Regression (GPR) model, the amplitude of the kernel function is controlled by the parameter σ_f, while the contribution of each input feature to the model prediction is determined by the length scale parameter l. During the model training phase, the objective is to fine-tune the model by optimizing a set of key hyperparameters $[\sigma_f, l, \sigma_n]$. This optimization process is typically accomplished by maximizing the model's marginal likelihood function, aiming to improve the model's fit to the data while maintaining its generalization ability. Given that GPR inherently adopts a non-parametric approach and can provide a probabilistic distribution of the predictions, it is particularly well-suited for modeling and predicting unknown nonlinear dynamic processes.

To compensate for the prediction errors of the Transformer model, two approaches were employed to establish residual models based on Gaussian Process Regression (GPR). One approach involved directly modeling the prediction errors of the Transformer model. The other method first utilized Empirical Mode Decomposition (EMD) to separate the original temperature series into two parts: the residual function and the Intrinsic Mode Functions (IMFs). Subsequently, the residuals representing the overall degradation trend of temperature were simulated through the Transformer model, while the remaining part (IMFs) was fitted by GPR. Table 2 lists the inputs and outputs of the aforementioned models, where Transformer I was trained using the original temperature series (y), and Transformer II used the temperature series residuals (Res) obtained from EMD. GPR I utilized the prediction errors (Δy) of Transformer I as its output, while GPR II considered y-Res as the target. To account for the effects of time, environmental conditions, and degradation on the local prediction errors of the battery core temperature, the input (X_2) of the residual models included three related features: ambient temperature, capacity degradation, and time steps.

TABLE 2. Model inputs and outputs.

Model	Input	Output
Transformer I	X_1	y
Transformer II	X_1	Res
GPR I	X_2	Δy
GPR II	X_2	y-Res

D. TRAINING

In this study, two distinct data sources, referred to as Dataset 1 and Dataset 2, were gathered to assess the effectiveness of the proposed temperature prediction method. The specifications of the experimental batteries are presented in Table 3. These datasets encompass batteries made from different materials, manufactured by Tianjin Lishen Battery Co., Ltd.: one set is composed of lithium iron phosphate (LiFePO4) batteries, and the other of lithium nickel cobalt aluminum oxide (LiNiCoAlO2) batteries, facilitating a comprehensive validation of the method's universality and reliability.

TABLE 3. Battery dataset information.

Parameters	Dataset 1	Dataset 2
Type	21700	18650
Materials	$LiFePO_4$	$LiNiCoAlO_2$
Rated Voltage	3.3	3.6
Rated Capacity	4.1	3
Voltage Range	2.0-3.7	2.5-4.2

Dataset 1 was examined in a laboratory setting, comprising three LiFePO4 batteries (designated as Battery 1, Battery 2, and Battery 3). All tests were conducted in a temperature-controlled chamber (RYHS100/3-40AB)

equipped with a cycler. To evaluate the environmental adaptability and reliability of the proposed method, mixed-driving cycle tests were performed on Batteries 1 through 3 at operational temperatures of 40°C, 10°C, and 30°C, respectively. These tests included the Urban Dynamometer Driving Schedule (UDDS) and the Federal Urban Driving Schedule (FUDS). During these procedures, the batteries were charged following a constant current-constant voltage (CC-CV) protocol at a 0.5C current (where C represents the nominal capacity), utilizing a high-precision battery testing system (CT-4008T). Data collection was managed by a data acquisition instrument (DAQ970A) connected to a computer, and specific dynamic configurations were maintained until the terminal voltage reached 2.0 V. Moreover, data at various stages of battery aging were acquired at a 1C discharge rate as the cycle count increased. Batteries 1, 2, and 3 experienced 598, 1770, and 1210 cycles, respectively, with their capacities declining from 4 Ah to 3.03 Ah, 3.56 Ah, and 1.65 Ah, in accordance with IEC (International Electrotechnical Commission) standards. Temperature monitoring was performed using T-type (copper-constantan) thermocouples, selected for their stability, high sensitivity, and broad measurement range, with a calibrated measurement error of less than ±0.15°C. The sampling frequency for each dataset was set at 1 Hz. Figures 7, 8, and 9 provide examples of the charge-discharge profiles of Battery 3, illustrating the phases of discharge, charge, rest, and discharge. Despite the operational temperature being set at 30°C, the surface temperature of the battery fluctuated rapidly with operation and significantly exceeded 30°C. This was attributed to the battery's brief rest period of only 10 minutes per cycle, preventing it from returning to the operational temperature during continuous cycling. Additionally, as the battery aged, its internal resistance increased, resulting in further temperature rises. Dataset 2 consisted of a LiNiCoAlO2 battery (designated as Battery 4). This battery

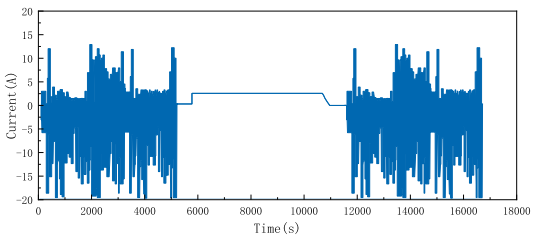


FIGURE 7. Battery 3 current file.

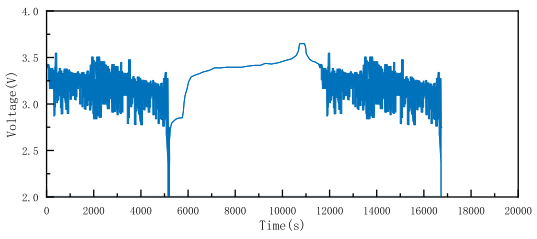


FIGURE 8. Battery 3 voltage file.

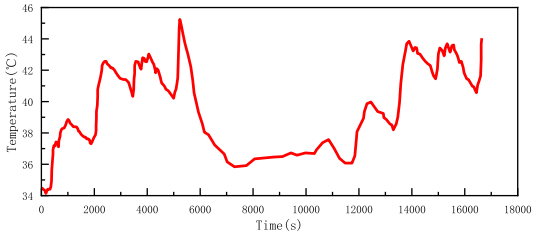


FIGURE 9. Battery 3 temperature variation.

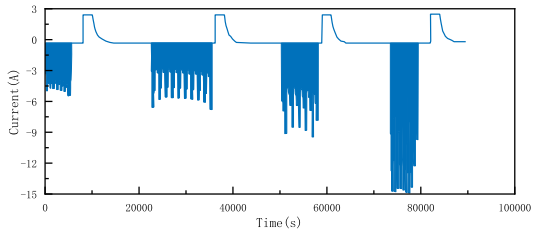


FIGURE 10. Battery 4 current file.

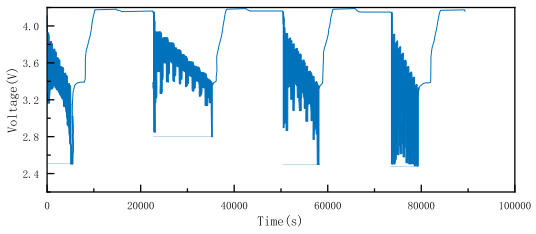


FIGURE 11. Battery 4 voltage file.

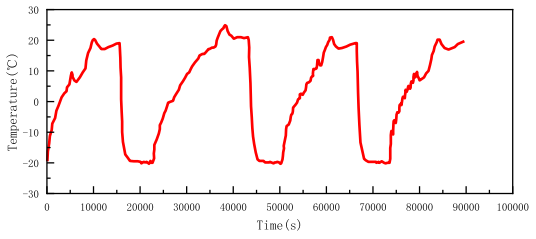


FIGURE 12. Battery 4 temperature variation.

was fully charged according to the CC-CV protocol and then discharged through various driving cycles, including UDDS, Los Angeles 92 cycle (LA92), Supplemental Federal Test Procedure driving cycle (US06), and the Highway Fuel Economy Driving Cycle (HWFET). Figures 10, 11, and 12 depict the current, voltage, and temperature profiles of Battery 4, indicating that the operating temperature fluctuated between −20°C and 20°C throughout the tests.

To determine the optimal architecture for application, hyperparameters were adjusted based on training data and independent test data. To prevent overfitting, the early stopping method was employed, meaning that the training would be halted if the mean squared error on the independent validation dataset increased for six consecutive time steps. For

this purpose, internal cross-validation within the training was utilized. The training dataset was divided into three subsets: 70% for training, 15% for validation, and 15% for testing. The mean squared error during the training of GPR and SVR depended on the type of residual model and occurred over multiple independent complete training cycles. Despite using the same dataset in each cycle, the training outcomes varied due to different random initialization values of weights at the start of training.

The training and testing results, along with the statistical errors in temperature estimation, are listed in Table 4. It was observed that both models were well-trained, exhibiting high accuracy with an average absolute error (MAE) below 1% for the test batteries. The MAE and Root Mean Square Error (RMSE) for both models did not exceed 1.3% and 1.7%, respectively, on the test batteries. Furthermore, the accuracy of the GPR model was slightly higher than that of the SVR model. These results validated that the selected feature set could adequately represent the variations in battery temperature.

TABLE 4. The statiscal error of battery temperature estimation.

	Way	MAE	RMSE
Training	SVR	0.97%	1.39%
	GPR	0.98%	1.27%
Test	SVR	1.21%	1.62%
	GPR	1.08%	1.40%

IV. RESULTS AND DISCUSSIONS

A. ANALYSIS OF MODEL ABLATION EXPERIMENTS

Batteries 1 and 2 from Dataset 1 were subjected to ablation experiments to evaluate various temperature prediction models. We calculated the mean absolute errors (MAE_ave) and root mean square errors (RMSE_ave) for each method, summarized in Table 5. All models demonstrated the ability to accurately predict battery temperature trajectories using early data inputs. Among them, the combination of Transformer I and Gaussian Process Regression (GPR) I outperformed others, achieving average absolute errors and root mean square errors of 1.18% and 1.47%, respectively. GPR I provided confidence intervals for estimated residuals, which were notably smaller than the temperature values themselves, enhancing the model's precision by about 12%.In contrast, Transformer II aimed to target the residuals of temperature sequences decomposed by Empirical Mode Decomposition (EMD); however, it did not yield significant improvements. The compensatory capability of GPR II was also less evident.

The performance of iterative and sequence-to-sequence prediction methods was further compared. Iterative methods utilize data from preceding steps to predict future temperatures; while these methods are accurate during training, they tend to accumulate errors during testing, thus lowering their predictive accuracy. Conversely, sequence-to-sequence methods maintained higher accuracy throughout testing phases,

TABLE 5. Statistical errors in battery temperature prediction under different models.

	Method	MAE_ave	RMSE_ave
Training	Transformer I	1.16%	1.57%
	Transformer I + GPR I	1.08%	1.45%
	Transformer II	1.26%	1.73%
	Transformer II + GPR II	1.24%	1.70%
	Iterative GPR	1.02%	1.34%
	Iterative LSTM	1.06%	1.44%
Testing	Transformer I	1.32%	1.64%
	Transformer I + GPR I	1.18%	1.47%
	Transformer II	1.31%	1.70%
	Transformer II + GPR II	1.29%	1.69%
	Iterative GPR	2.33%	3.35%
	Iterative LSTM	2.55%	3.24%

as shown in Table 5, making them more reliable for predicting battery temperature series.

A sensitivity analysis of the model was conducted using the initial aging dataset of Battery 4, which is an 18650-type lithium battery. To validate the effectiveness of the feature selection procedure, the prediction results based on different feature sets were compared. The three sets of features, shown in Table 6, were used as inputs for the Transformer-GPR model. Feature set N1, determined by the proposed feature factors, includes variables such as time step, discharge current, SOC, voltage (U), heat generation rate (Q), capacity loss (Closs), and current temperature. Feature set N2 excludes the capacity decay compensation, while N3 omits the heat generation rate input. The results based on these feature sets are shown in Figure 13. Additionally, the average Mean Absolute Error (MAE) and average Root Mean Square Error (RMSE) obtained from the experiments are listed in Table 6. It can be observed that N1 achieved the highest accuracy. Due to the absence of heat generation rate and capacity decay inputs, N2 and N3 exhibited an increase in prediction error by 0.25% and 0.12% in average MAE compared to the N1 dataset, respectively. Therefore, feature set N1 is identified as the optimal choice.

TABLE 6. Comparison of feature set evaluation metrics.

Characteristic factor	MAE_ave	RMSE_ave
N1	1.20%	1.61%
N2	1.45%	1.92%
N3	1.32%	1.67%

B. BATTERY PREDICTION COMPARATIVE STUDY

In this section, the accuracy and reliability of the proposed method at various ng temperatures were first verified, and the prediction results were compared with those of different algorithms. Subsequently, the universality of the proposed method was evaluated based on different types of batteries.

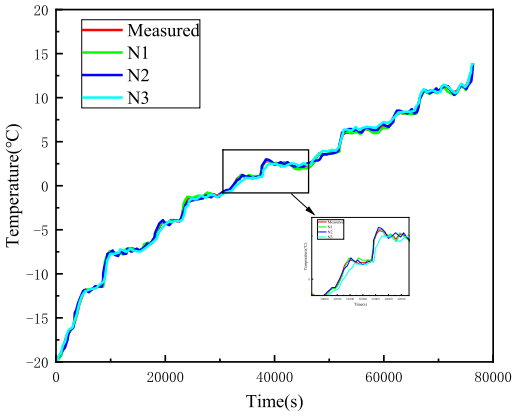


FIGURE 13. Temperature variations for different feature sets.

It is noteworthy that the ratio of the model training set to the test set was 7:3. Moreover, for the convenience of evaluating the model's prediction outcomes, Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Maximum Error (MAX) were employed as metrics for assessing the model's predictive performance. RMSE, a common metric for evaluating predictive models of continuous variables, measures the square root of the average of the squares of the differences between predicted and actual values. MAE measures the average of the absolute differences between predicted and actual values, providing a simple measure of predictive accuracy that assigns equal weight to all sizes of errors. MAX error measures the largest difference among all predicted and actual values. The definitions of these three evaluation metrics are as follows:

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2} \tag{34}$$

$$MAE = \frac{1}{n}\sum_{i=1}^{n}\left|y_i - \hat{y}_i\right| \tag{35}$$

$$MAX = max\left(\left|y_i - \hat{y}_i\right|\right) \tag{36}$$

Among them, y_i is the i-th actual value, $\hat{y}_i$ is the i-th predicted value, and n is the number of observed values.

The influence of the environment on temperature prediction being significant, the proposed Transformer-GPR network was first validated using Dataset 1, collected at various ambient temperatures. In this paper, the leave-one-out method is employed to assess our model: a battery is randomly selected for testing, while the remaining batteries are utilized for training. Following the test iterations, the average score across all batteries is calculated. It is important to note that this part utilized initial aging data from Batteries 1 to 3. Initially, the data from Battery 1 at an ambient temperature of 40°C was used for validation, with the prediction results shown in Figures 14 and 15. It was clearly observed that despite the experiments being conducted at 40°C, the surface temperature of the battery exhibited a fluctuating increase, attributed to the charging and discharging of the battery, introducing uncertainty and unpredictability to temperature prediction. Fortunately, the proposed method's battery temperature prediction curve was able to effectively track the actual temperature curve, with errors within ±0.3°C.

Subsequently, predictions using data from Battery 2 at 10°C were conducted, with results depicted in Figures 16 and 17. Despite the surface temperature of

TABLE 7. The comparison of different battery temperature prediction methods.

Method	Temperature	Battery type	RMSE	MAX	MAE
My Model	40℃ (battery 1)	LiFePO4	0.046	0.26	0.037
	10℃ (battery 2)		0.10	0.57	0.084
	30℃ (battery 3)		0.09	0.56	0.081
ARIMA	—		0.15	0.62	0.11
SVM	—		0.18	1.17	0.13
ET	—		0.25	1.94	0.17
BPNN	—		0.39	2.26	0.28
GRU[27]	25℃	LiFePO4	0.24	1.60	0.18
LSTM[28]	—	INR18650-35E	0.16	0.66	0.13
ANFIS[29]	—	$LiNiCoAlO_2$	0.14	0.61	0.15
NARX[16]	25℃	LiFePO4	0.15	—	—
EIS-EKF[30]	8℃	LiFePO4	1.35	—	—

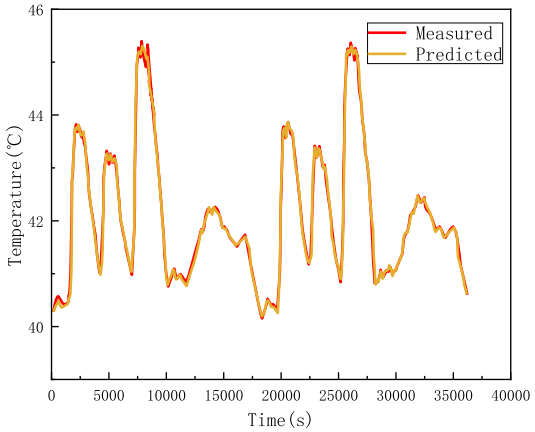


FIGURE 14. Battery 1 temperature variation.

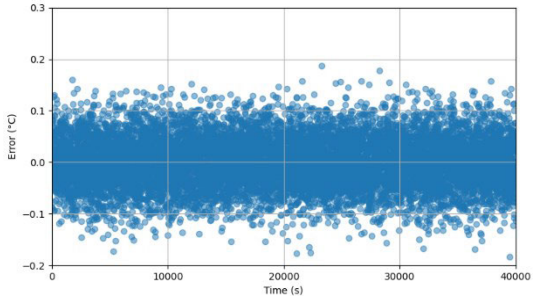


FIGURE 15. Battery 1 prediction error distribution.

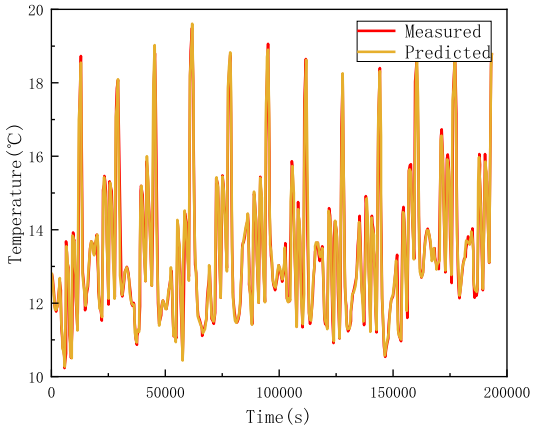


FIGURE 16. Battery 2 temperature variation.

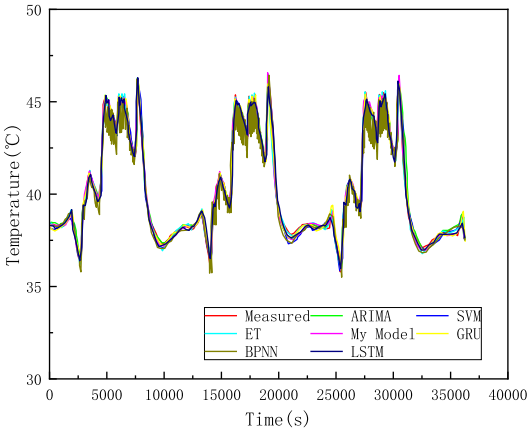


FIGURE 18. Temperature variation of battery 3 using different methods.

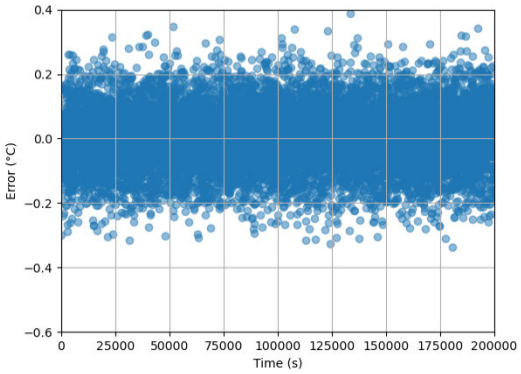


FIGURE 17. Battery 1 prediction error distribution.

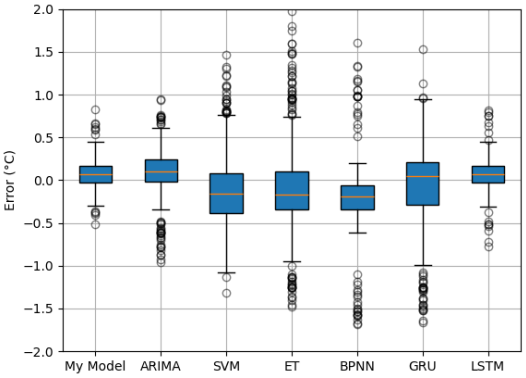


FIGURE 19. Temperature prediction error of battery 3.

Battery 2 varying between 10°C and 20.2°C, the predicted curve of the proposed method closely matched the actual temperature curve. As shown in Table 7, the RMSE, MAX, and MAE for Battery 2 were 0.10°C, 0.57°C, and 0.084°C, respectively, demonstrating the predictive efficacy of the proposed algorithm.

Next, initial aging data from Battery 3 at a working environment of 30°C was used to validate the proposed method. To further evaluate the performance of the proposed method, six common machine learning algorithms were compared, including Autoregressive Integrated Moving Average (ARIMA), Support Vector Machine (SVM), Extra Trees (ET), BP Neural Network (BPNN), Gated Recurrent Unit (GRU), and Long Short-Term Memory (LSTM). Detailed comparison results of the different algorithms are shown in Figures 18 and 19. It is evident from the figures that all comparative methods exhibit significant fluctuations in their temperature prediction curves, indicating difficulties in accurately tracking the temperature change trajectory. Additionally, the error distribution of the proposed method is more concentrated and closer to zero, whereas the errors from other methods are more dispersed. This was because traditional machine learning methods, including ARIMA, SVM, ET, and BPNN, only consider current measurement signals as inputs, failing to adaptively adjust to changes in historical information. Moreover, GRU and LSTM have limitations in processing long sequences and parallelization. In contrast, the proposed model in this paper utilized a Transformer model to predict the future temperature trajectory of the battery and further introduced two residual models to compensate for prediction errors caused by local temperature changes. The Transformer model demonstrated its effectiveness in capturing the overall trend of battery temperature changes, while the Gaussian Process Regression (GPR) model successfully compensated for local temperature variations.

For a more intuitive comparison of the estimation results, Table 7 lists the RMSE, MAX, and MAE for all methods. It was observed that the temperature prediction performance of the proposed method was superior to all comparative methods, with BPNN showing the poorest performance among the candidates. Specifically, the RMSE values for ARIMA, SVM, ET, BPNN, GRU, and LSTM were 0.15°C, 0.18°C, 0.25°C, 0.39°C, 0.24°C, and 0.16°C, respectively, while the RMSE for the proposed method was only 0.09°C. Additionally, the MAX and MAE for the proposed method were 0.56°C and 0.081°C, respectively, significantly lower than those of the six comparative methods. Furthermore, temperature prediction results reported in the literature were

compared with the proposed method, showing statistical errors in Table 7. In [16], a Nonlinear Autoregressive with Exogenous Inputs (NARX) network was used to estimate battery temperature, where the smallest RMSE reported was 0.15°C, considerably larger than the RMSE for the proposed method. In summary, the comparison results demonstrated the satisfactory accuracy of the proposed method and confirmed its superiority and reliability in temperature prediction.

C. TEMPERATURE PREDICTION AT DIFFERENT CONDITIONS

The Transformer-GPR has shown improvement in generalization capability compared to data-driven methods, illustrated through two cases. The first case involved testing the operating conditions of batteries in the test set falling within the distribution range of the training data. Subsequent experiments were conducted on different batteries, albeit under operating conditions identical to those in the training data. The test set for these experiments was randomly selected from a dataset of lithium-ion batteries with common environmental conditions (15°C to 35°C) and typical discharge rates (1C to 3C).

TABLE 8. Mean squared error (MSE) of different methods under different conditions.

Operating conditions	GRU	LSTM	Transformer-GPRI	Transformer-GPRII
Charge 0.5C, Discharge 2C,35°C	0.253	0.143	0.102	**0.058**
Charge 0.5C, Discharge 3C,25°C	0.213	0.211	0.236	**0.117**
Charge 0.5C, Discharge 1C,15°C	0.338	0.285	0.251	**0.149**

Mean Squared Error (MSE) between the predicted values and the experimental measurements was used to assess the accuracy of the predictions, with the results listed in Table 8. For comparison, GRU and LSTM were configured with three layers of 120 neurons each, while both Transformer-GPRI and Transformer-GPRII were set up with three-layer encoders. The results indicated that Transformer-GPRII provided more accurate predictions compared to existing methods.

The second case involved testing conditions of the batteries in the test set exceeding the distribution range of the training set data. A decline in the predictive accuracy of all methods was observed. Thus, under these circumstances, Transformer-GPRII outperformed other data-driven methods. Further comparative experiments between Transformer-GPRII and LSTM were conducted on batteries under the same operating conditions as the training data. During training, data from charging at 0.5C and discharging at 2C under an ambient temperature of 25°C were used. Another dataset involved testing at 0.5C charging and 3C discharging under the same ambient temperature. Figure 20(a) showcased the predictive performance of Transformer-GPR, and Figure 20(b) displayed that of LSTM. Since the training data was based on 2C discharging, LSTM predicted lower temperatures under 3C discharging conditions, whereas Transformer-GPRII demonstrated better predictive outcomes. Their MSEs on the test set were 1.075 and 2.778, respectively.

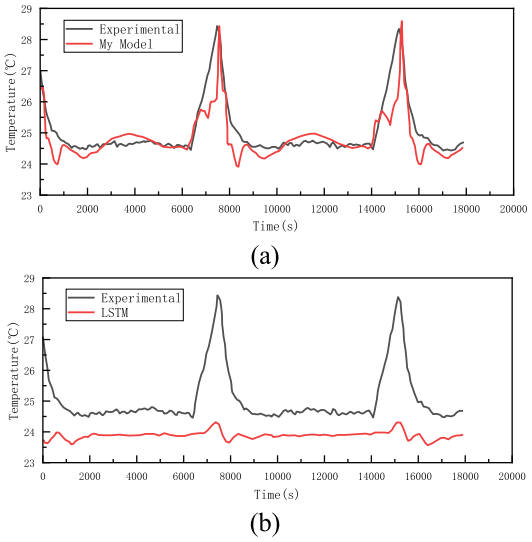


FIGURE 20. My model and LSTM temperature prediction performance.

In practice, with the changing real-world operating environments, new data would be used for incremental training. Transformer-GPR maintained good transferability. When shifted to entirely different operating conditions not present in the previous training set, the pre-trained Transformer-GPRII required only a small amount of data under new operating conditions for incremental training to accurately predict temperature. Experiments were conducted as follows: Transformer-GPRII and LSTM were pre-trained with complete lifecycle data at 0.5C charging and 3C discharging under an ambient temperature of 25°C. To predict battery temperature under 35°C ambient temperature at 0.5C charging and 1C discharging conditions, both models were incrementally trained with the first 30 cycles of new battery data. Figures 21(a) and (b) showed the predictive results of Transformer-GPR and LSTM from 30 to 230 cycles, with their MSEs being 0.026 and 15.021, respectively. It was evident that, as shown in Figure 21(c), LSTM performed well in predicting over shorter intervals in later cycles, but the subsequent predictive accuracy significantly decreased, as depicted in Figure 21(d), due to overfitting caused by insufficient data. Since Transformer-GPRII provides a flexible attention

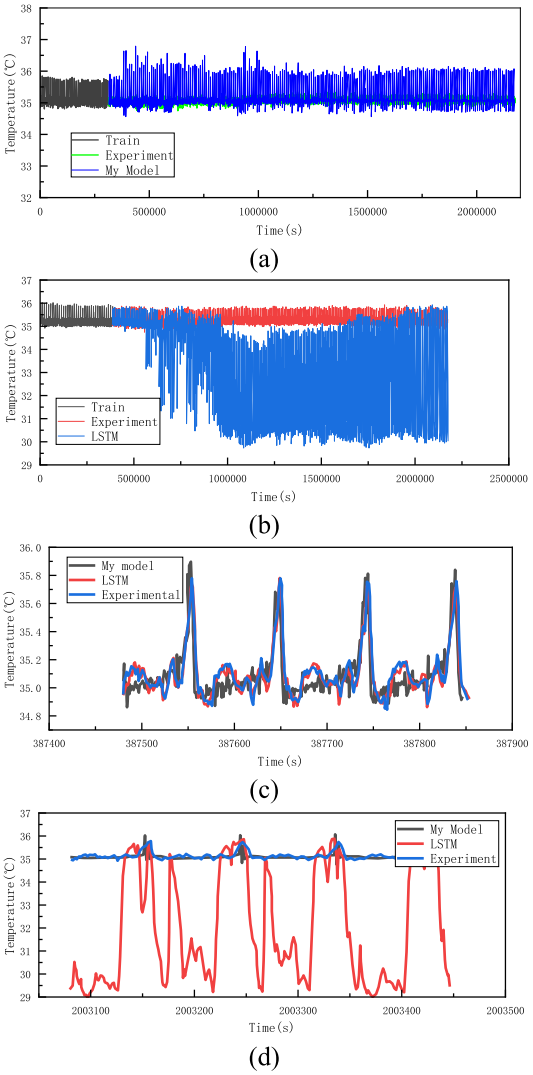


FIGURE 21. At 35°C, using a minimal dataset for training at 0.5C charge and 1C discharge rates, (a) and (b) demonstrate the predictive abilities of My Model versus LSTM for upcoming cycles, with (c) showing initial predictions and (d) illustrating performance after 210 cycles.

mechanism integrated with Gaussian Process Regression (GPR), capable of effectively compensating for prediction errors due to local temperature changes while avoiding overfitting, it consistently maintained good long-term predictive performance.

D. REAL VEHICLE GENERALIZATION VERIFICATION

To verify the generalization capability of the proposed model on actual vehicles equipped with a Battery Thermal Management System (BTMS), a pure electric SUV was selected as the test subject, with data from similar vehicles used as the offline training set. The experiment aimed at predicting the battery's average temperature was conducted by simulating driving conditions under the China Light-Duty Vehicle Test Cycle (CLTC), as illustrated in Figure 22. A serpentine surface liquid cooling plate was the BTMS employed. The experimental results, showing a maximum temperature

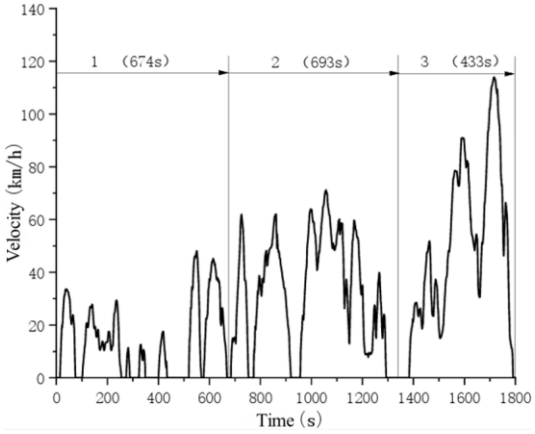


FIGURE 22. CLTC vehicle conditions [31].

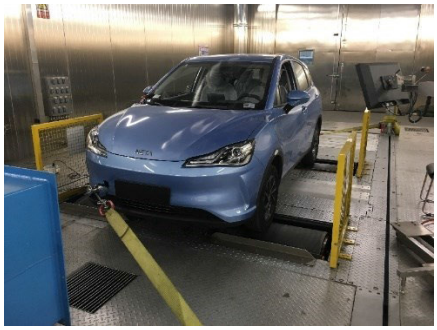


FIGURE 23. Vehicle dynamometer schematic diagram.

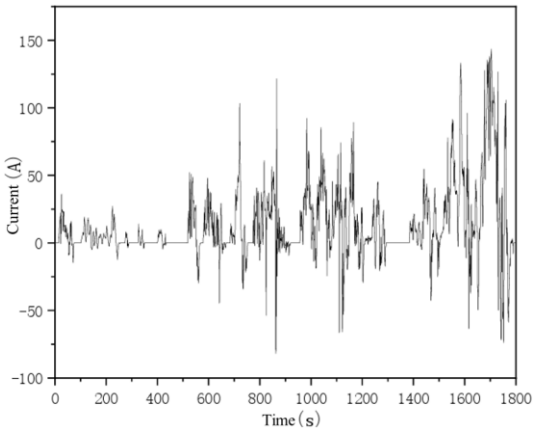


FIGURE 24. Dynamic test condition current curve.

difference between batteries of less than 0.8K, suggest that the battery pack can be treated as a unified whole. Parameters related to the liquid cooling plate, such as coolant temperature and flow rate, were also incorporated in the experiment to adequately consider the impact of the liquid cooling plate on battery temperature. Conducted on a specialized test bench as shown in Figure 23, the vehicle was placed on a dynamometer to simulate CLTC conditions, and data required for model input, like the measured current shown in Figure 24, were

collected. Comparison of predicted temperatures with actual measurements yielded an RMSE of 0.048, an MAE of 0.036, and a MAX of 0.28, indicating high precision and excellent generalization capability of the model, starting from 650 time steps.

The model proposed in this study not only fits the test data accurately but also demonstrates comparable performance in real-world application scenarios, proving its strong generalization ability. Such high precision and generalization performance are crucial for the electric vehicle industry, ensuring battery stability and safety under varying environmental and operational conditions. Moreover, the comprehensive consideration of BTMS parameters in the model further enhances its practical value, making it a powerful tool for the design and optimization of thermal management systems in electric vehicles.

V. CONCLUSION AND PROSPECT

This study introduces a novel battery temperature prediction method, Transformer-GPR, which effectively addresses the challenges of temperature prediction throughout the battery's lifecycle by utilizing battery charge and discharge data along with data-driven algorithms. The method employs a Transformer model to analyze key features and predict future temperature sequences, while residual models based on Gaussian Process Regression (GPR) adjust for prediction errors. When compared with six common machine learning algorithms and various new methods in the literature, such as LSTM, EKF, and BPNN, the Transformer-GPR II model demonstrated the highest prediction accuracy, achieving the lowest values for RMSE, MAX, and MAE, at 0.046, 0.26, and 0.037, respectively. Additionally, the execution time for each prediction did not exceed 6.8 milliseconds, highlighting its potential for online battery temperature prediction applications. To further validate its effectiveness, experiments were conducted using the China Light Duty Vehicle Test Cycle (CLTC) dataset collected from actual electric vehicles. The results confirmed the superior performance of the Transformer-GPR II model.

Despite the impressive performance of the Transformer-GPR framework on existing datasets, scalability challenges arise when applied to different scales, particularly within electric vehicles and large battery arrays, where increased data volumes and computational demands are inevitable. Future research should focus on optimizing training efficiency and inference speed for large datasets without compromising model performance. Techniques such as distributed training frameworks or more efficient parallel computing algorithms are essential for enhancing model scalability.

Furthermore, although the framework exhibits robust performance under controlled experimental conditions, significant environmental variations in real-world applications, such as temperature fluctuations and changes in current load, demand higher model robustness. To enhance adaptability, integrating a greater variety of training data reflecting different environmental conditions, employing domain adaptation techniques such as adversarial training, and developing online learning mechanisms for dynamic model adjustment based on new data are crucial strategies for future work.

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